

Database Benchmark Report

Automatically generated benchmark report

Date: 2026-07-02 14:08:29

Operating System: Linux

Python: 3.12.3

Databases: H2, HSQLDB

Benchmarks: 8

Measured Runs: 20

Warm-up Runs: 5

Benchmark Summary

Number of databases: **2**

Benchmark categories: **8**

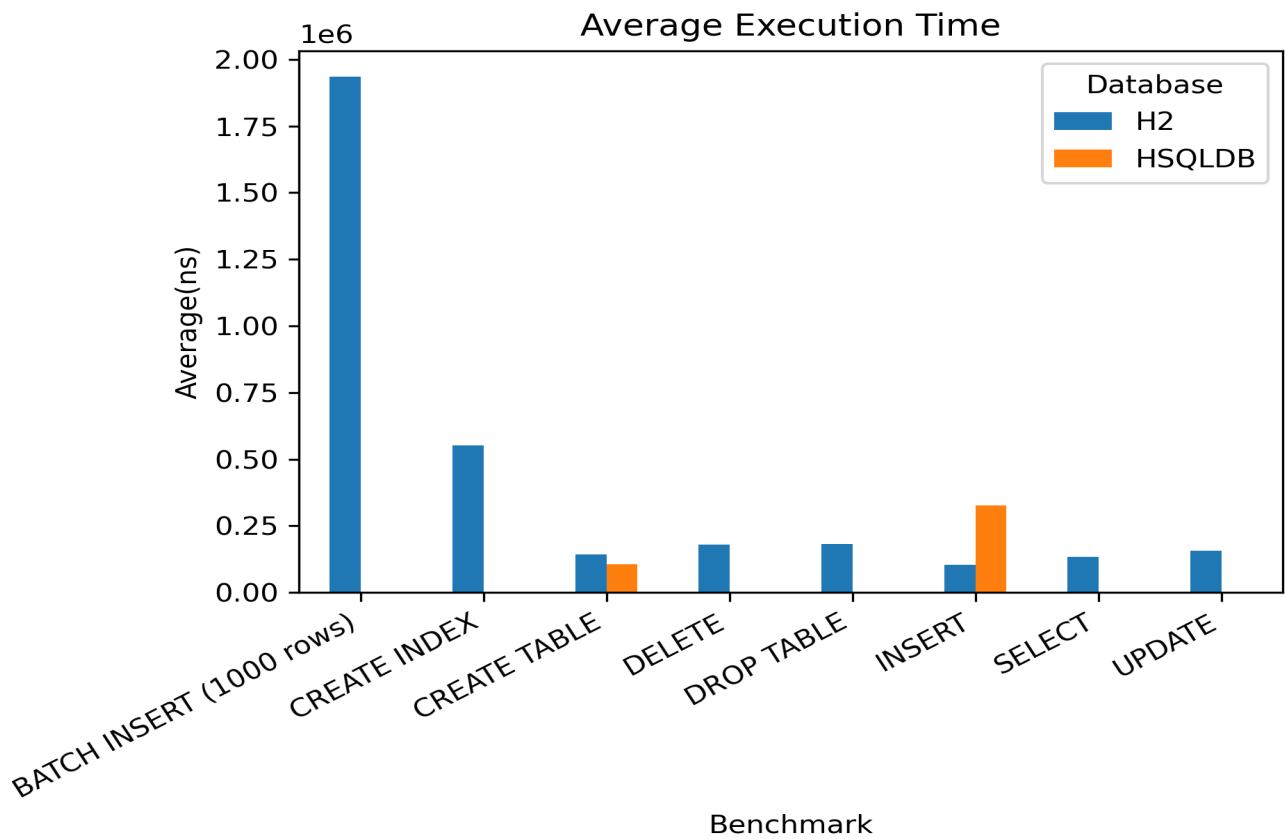
Benchmark results: **10**

This report summarizes execution-time measurements for embedded relational database management systems. All measurements were performed after JVM warm-up iterations and repeated multiple times to improve statistical reliability.

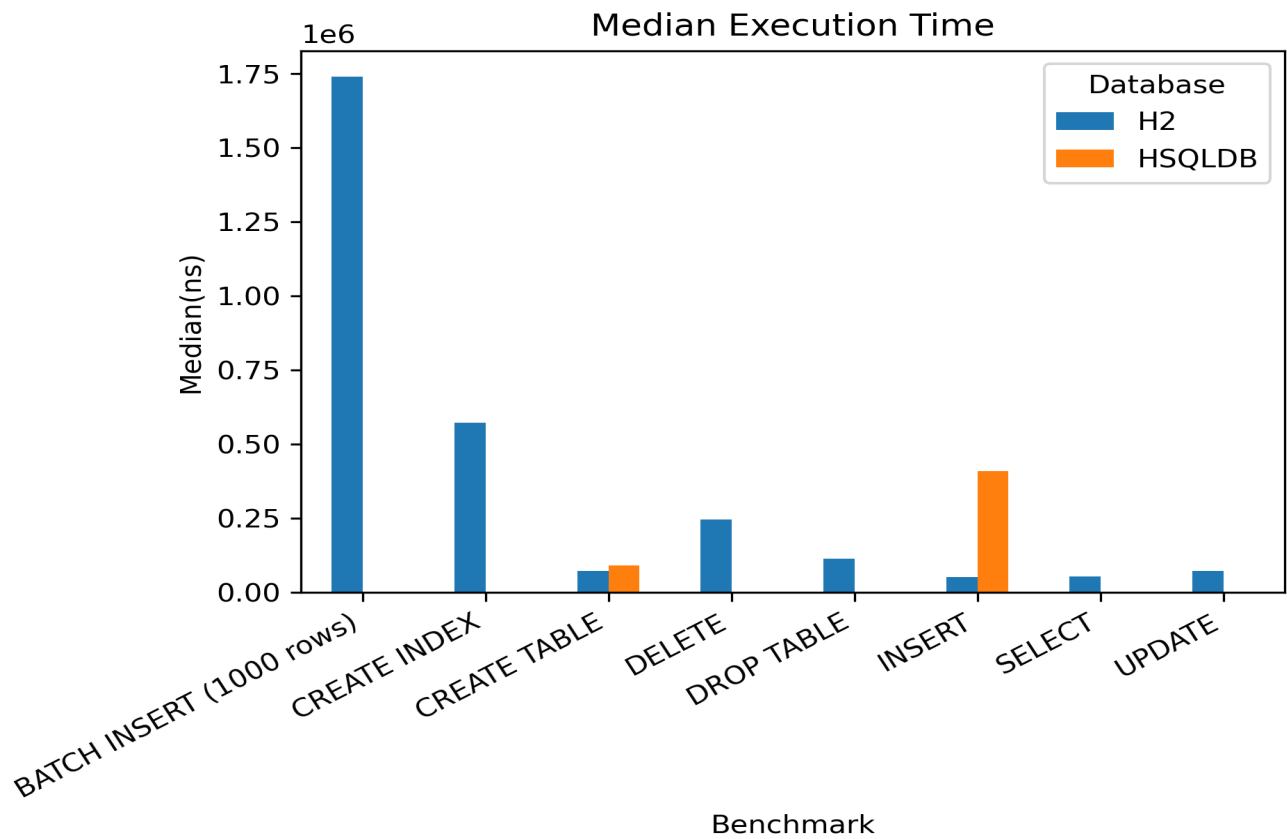
Summary Statistics

Database	Benchmark	Average	Median	Std Dev	Minimum	Maximum
H2	CREATE TABLE	141651	72217	131781	60545	6206964
H2	INSERT	103631	51227	101291	42951	7032246
H2	BATCH INSERT (1000 rows)	1935152	1738870	1414747	1348201	40959219
H2	SELECT	134216	54122	117198	43513	4415865
H2	UPDATE	156059	72557	144978	44744	25642948
H2	DELETE	180206	245514	110257	45175	2839402
H2	DROP TABLE	181378	112984	133247	51848	5448047
H2	CREATE INDEX	552419	572353	273023	343470	11631658
HSQldb	CREATE TABLE	105320	91012	104485	41038	19665276
HSQldb	INSERT	327234	408336	189230	113895	9004315

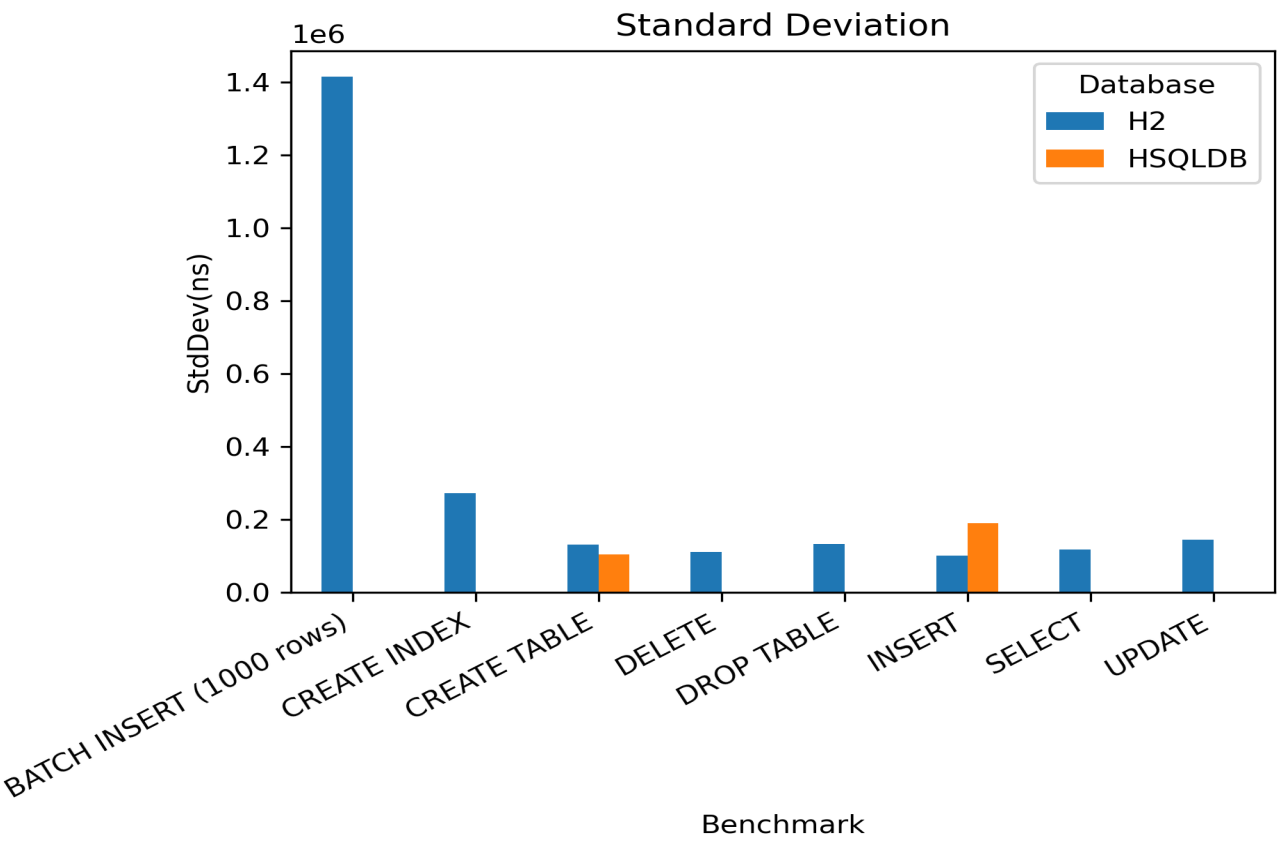
Average Execution Time



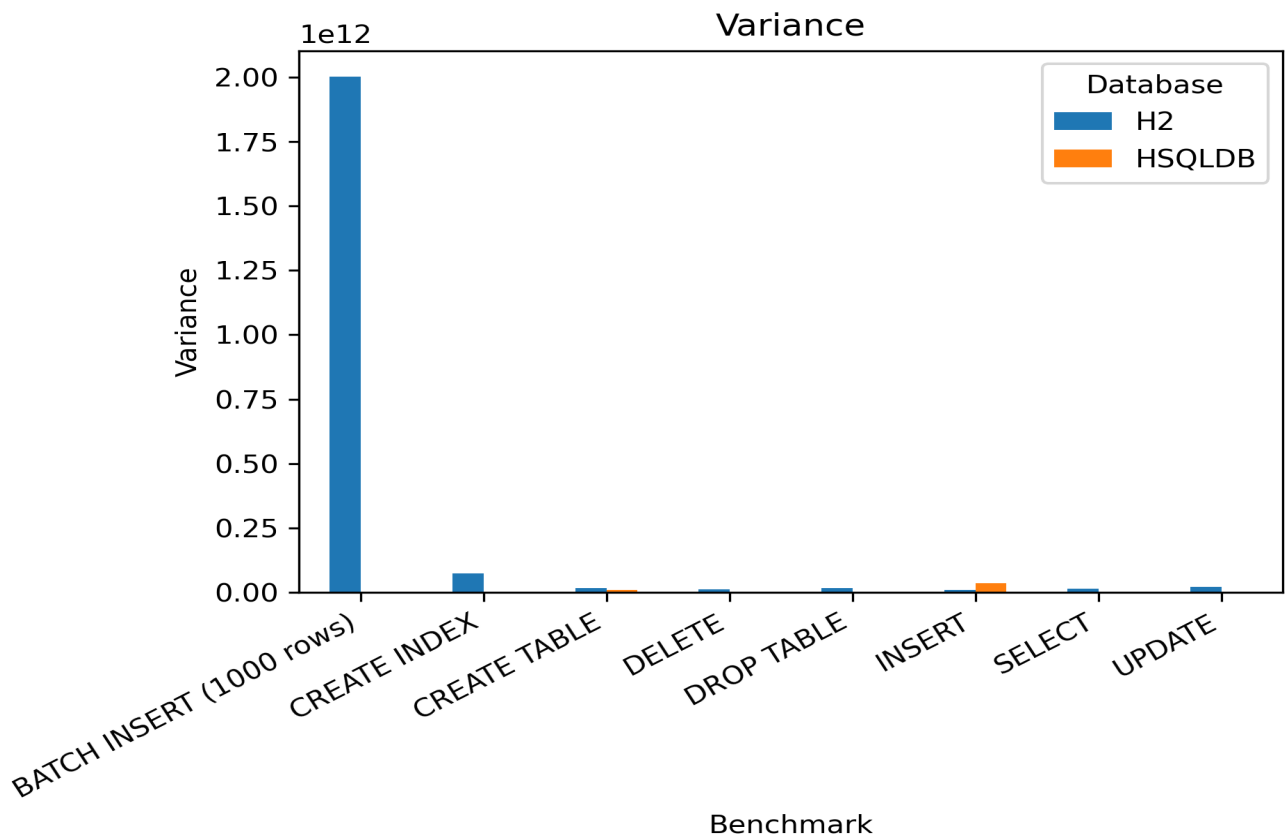
Median Execution Time



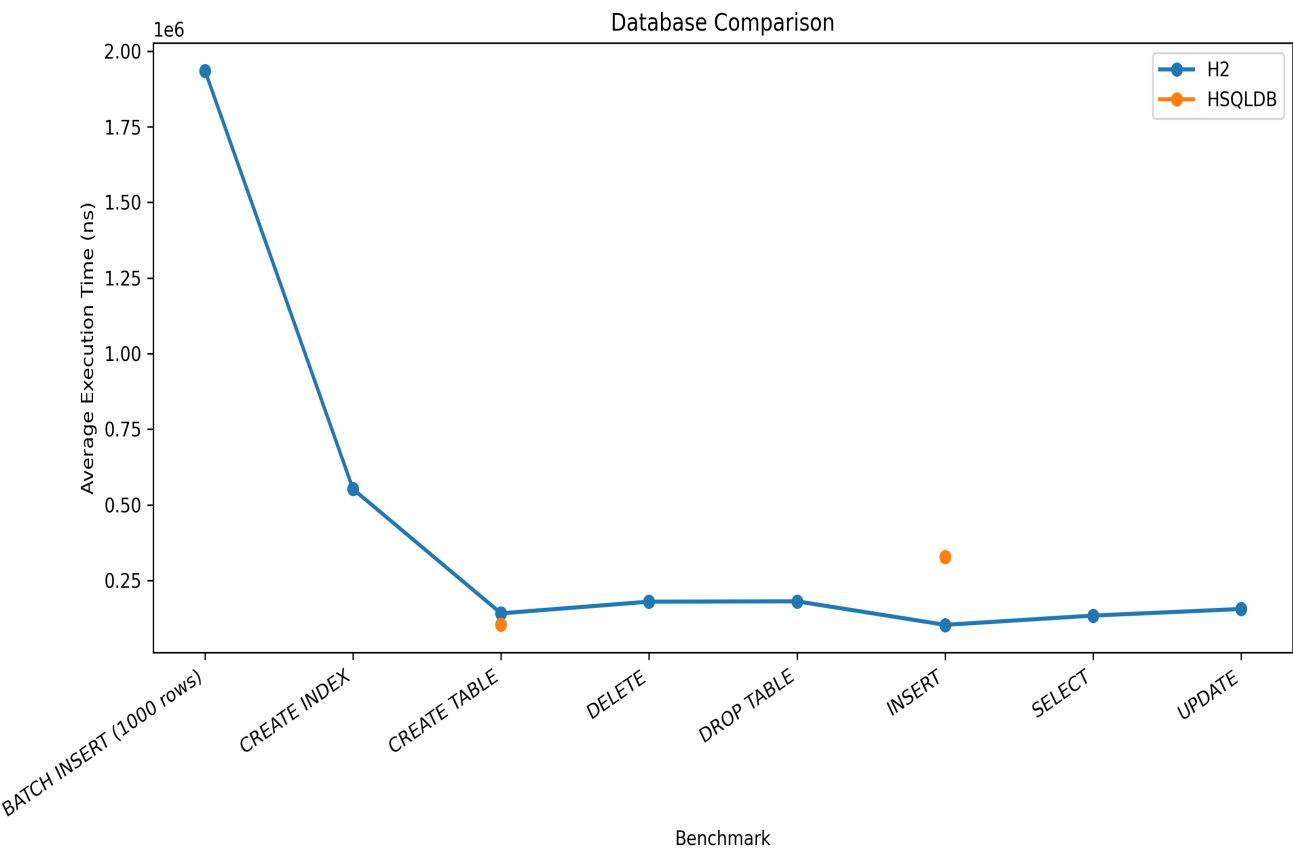
Standard Deviation



Variance



Database Comparison



Benchmark Winners

Benchmark	Winner	Average (ns)
BATCH INSERT (1000 rows)	H2	1,935,152
CREATE INDEX	H2	552,419
CREATE TABLE	HSQLDB	105,320
DELETE	H2	180,206
DROP TABLE	H2	181,378
INSERT	H2	103,631
SELECT	H2	134,216
UPDATE	H2	156,059

Overall Benchmark Score

Database	Benchmarks Won
H2	7
HSQLDB	1

Student's t-test

Benchmark	t-statistic	p-value	Significant?
BATCH INSERT (1000 rows)	nan	nan	NO
CREATE INDEX	nan	nan	NO
CREATE TABLE	68.316	0.000000	YES
DELETE	nan	nan	NO
DROP TABLE	nan	nan	NO
INSERT	-329.439	0.000000	YES
SELECT	nan	nan	NO
UPDATE	nan	nan	NO

Interpretation

A p-value below 0.05 indicates that the observed performance difference between H2 and HSQLDB is statistically significant.

Automatic Conclusion

Summary

The benchmark suite executed 8 benchmark categories on 2 embedded database systems.

Based on the average execution time, **H2** achieved the best overall performance, winning 7 benchmark categories.

The fastest benchmark measured was **INSERT** executed on **H2** with an average execution time of **103,631 ns**.

The slowest benchmark measured was **BATCH INSERT (1000 rows)** executed on **H2** with an average execution time of **1,935,152 ns**.

Overall, the benchmark results indicate that execution times remained stable after warm-up iterations while demonstrating measurable performance differences between the tested database systems.

This report was generated automatically from the benchmark CSV produced by the Java benchmark framework.